

Real-Time Cybersecurity Event Analytics Platform

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1. Idea / Problem Statement

Modern applications, servers, authentication systems and network devices continuously generate large amounts of security-related data, including login attempts, API requests, IP addresses and network activity. With such data being generated continuously and from multiple sources, manually analyzing or processing these events after they have been collected can make it difficult to identify unusual or potentially harmful activity in a timely manner.

The proposed project aims to develop a **Real-Time Cybersecurity Event Analytics Platform** that can continuously collect, process and analyze simulated security events from different sources. The system will monitor incoming activity and identify potentially suspicious patterns, such as repeated failed login attempts, unusual traffic behavior, or abnormal access patterns. Along with real-time analysis, the system will maintain event data for further historical analysis and understanding of security trends.

The platform will also provide an interactive environment for generating different types of normal and abnormal security activity, allowing the complete flow of events and their analysis to be demonstrated in real time. A monitoring interface will present ongoing events, detected anomalies, alerts, and relevant system statistics. The project will thus explore how data engineering and real-time analytics can be applied to a practical cybersecurity problem.

2. Objectives

The major objectives of the project are:

- To develop a system for handling continuously generated security-related events from multiple sources.
- To enable real-time processing and analysis of incoming event data.
- To identify unusual patterns and potentially suspicious security activity.
- To maintain event information for both real-time monitoring and historical analysis.
- To provide meaningful security insights through visualizations and analytics.
- To create a controllable environment for simulating different types of security activity.
- To explore the use of AI-based techniques to assist in understanding and summarizing detected security events

3. Planned Functionalities

The project is broadly planned to include the following functionalities:

- **Security Event Simulation:** Generate realistic events from sources such as authentication systems, applications, servers, and network devices, including both normal and abnormal activity.
- **Real-Time Event Processing:** Continuously receive and process incoming events while handling common challenges associated with continuously generated data.
- **Anomaly and Pattern Detection:** Analyze event streams to identify suspicious or unusual behavior and generate relevant alerts.
- **Data Storage and Historical Analysis:** Preserve event data and processed information to support analysis of past activity, trends, and recurring patterns.
- **Real-Time Monitoring Dashboard:** Provide a visual interface showing current events, detected anomalies, alerts, trends, and overall system activity.
- **Interactive Simulation Interface:** Allow users to trigger different scenarios and observe how the system responds as events move through the platform.
- **System Monitoring:** Provide visibility into the health and performance of the data processing pipeline.

4. Data Engineering Concepts

The project will explore the following data engineering concepts:

- **ETL/ELT Data Pipelines:** For collecting, processing, transforming, and storing security event data.
- **Real-Time Stream Processing Architectures:** For continuously processing incoming events and identifying suspicious activity with low latency.
- **Data Lake Architecture:** For retaining raw security event data for future investigation, reprocessing, and historical analysis.
- **Data Warehouse Architectures:** For storing processed and aggregated information to support historical analytics and reporting.
- **Analytics & Monitoring Systems:** For visualizing security events, detected anomalies, trends, and the health of the data processing pipeline.